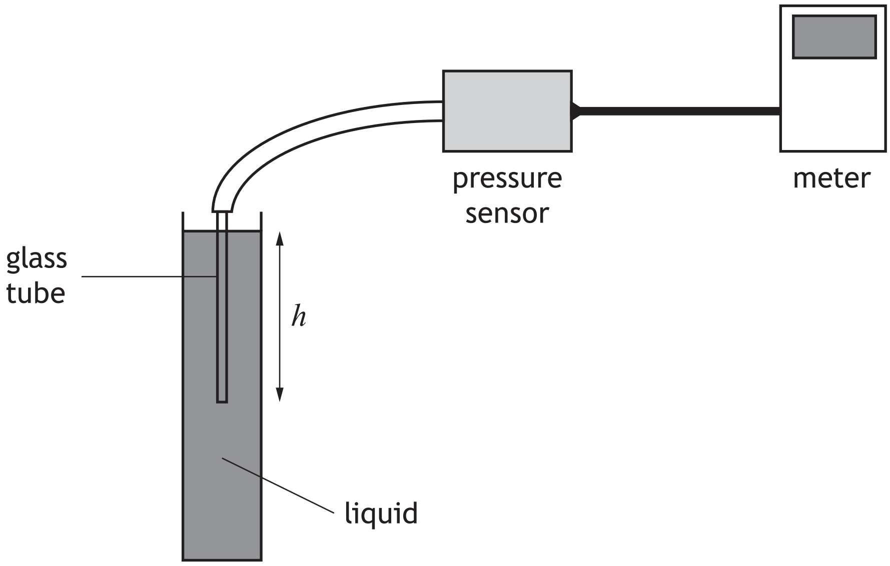
MARKS | DO NOT WRITE IN THIS MARGIN

13. A student sets up an experiment to investigate the pressure due to a liquid as shown.



The pressure due to a liquid is given by the relationship

$$p = \rho g h$$

where p is the pressure due to the liquid in pascals (Pa),

g is the gravitational field strength in N kg^{-1},

ρ is the density of the liquid in kg m^{-3},

and h is the depth in the liquid in m.

(a) The student initially carries out the investigation using water.

The density of water is $1{\cdot}00 \times 10^3\ \text{kg m}^{-3}$.

Calculate the pressure due to the water at a depth of 0·35 m. **2**

Space for working and answer

13. (continued)

(b) The student repeats the experiment with a different liquid.

The pressure meter is set to zero before the glass tube is lowered into the liquid.

The student takes measurements of the pressure at various depths below the surface of the liquid.

The student records the following information.

Depth, h (m)	Pressure, p (kPa)
0·10	1·2
0·20	2·5
0·30	3·6
0·40	4·9
0·50	6·2

(i) Using the square-ruled paper on *page 43*, draw a graph of p against h. **3**

(Additional graph paper, if required, can be found on *page 44*.)

(ii) Calculate the gradient of your graph. **2**

Space for working and answer

(iii) Determine the density of this liquid. **2**

Space for working and answer

[END OF QUESTION PAPER]